

Program 9:

Project Structure:

Pr9

-> namespaces.yaml

-> pods.yaml

Step 1: Create namespaces.yaml

```
! namespaces.yaml ×
! namespaces.yaml
1  apiVersion: v1
2  kind: Namespace
3  metadata:
4    name: dev
5  ---
6  apiVersion: v1
7  kind: Namespace
8  metadata:
9    name: prod
10
```

Step 2: Create pods.yaml

```
! pods.yaml
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: frontend-pod
5    namespace: dev
6    labels:
7      app: frontend
8      env: dev
9  spec:
10   containers:
11     - name: frontend
12       image: nginx
13       ports:
14         - containerPort: 80
15   ---
16   apiVersion: v1
17   kind: Pod
18   metadata:
19     name: backend-pod
20     namespace: prod
21     labels:
22       app: backend
23       env: prod
24   spec:
25     containers:
26       - name: backend
27         image: nginx
28         ports:
29           - containerPort: 80
30
```

Step 3: Apply namespaces and pods

```
kunaltiwari@riot pr9 % kubectl apply -f namespaces.yaml
```

```
namespace/dev created
```

```
namespace/prod created
```

```
kunaltiwari@riot pr9 % kubectl apply -f pods.yaml
```

```
pod/frontend-pod created
```

```
pod/backend-pod created
```

Step 4: Verify pods in each namespace

```
kunaltiwari@riot pr9 % kubectl get pods --all-namespaces
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
dev	frontend-pod	1/1	Running	0	51s
kube-system	coredns-66bc5c9577-thphh	1/1	Running	2 (82s ago)	10d
kube-system	coredns-66bc5c9577-v4hj2	1/1	Running	2 (82s ago)	10d
kube-system	etcd-minikube	1/1	Running	2 (82s ago)	10d
kube-system	kube-apiserver-minikube	1/1	Running	2 (82s ago)	10d
kube-system	kube-controller-manager-minikube	1/1	Running	2 (10d ago)	10d
kube-system	kube-proxy-dfzml	1/1	Running	2 (10d ago)	10d
kube-system	kube-scheduler-minikube	1/1	Running	2 (10d ago)	10d
kube-system	storage-provisioner	1/1	Running	5 (41s ago)	10d
prod	backend-pod	1/1	Running	0	51s

Step 5 :Filter pods by label

```
kunaltiwari@riot pr9 % kubectl get pods -n dev -l app=frontend
```

NAME	READY	STATUS	RESTARTS	AGE
frontend-pod	1/1	Running	0	88s

```
kunaltiwari@riot pr9 % kubectl get pods -n prod -l app=backend
```

NAME	READY	STATUS	RESTARTS	AGE
backend-pod	1/1	Running	0	95s

```
kunaltiwari@riot pr9 % █
```

Step 5 :Cleanup Process

```
kunaltiwari@riot pr9 % kubectl delete pods --all
```

```
pod "nginx-deployment-7cccd94f7-5qp8f" deleted from default namespace
```

```
pod "nginx-deployment-7cccd94f7-gzm2v" deleted from default namespace
```

```
kunaltiwari@riot pr9 % kubectl delete svc --all
```

```
service "kubernetes" deleted from default namespace
```

```
service "nginx-service" deleted from default namespace
```

```
kunaltiwari@riot pr9 % kubectl delete deployments --all
```

```
deployment.apps "nginx-deployment" deleted from default namespace
```

```
kunaltiwari@riot pr9 % kubectl delete namespace dev prod
```

```
namespace "dev" deleted
```

```
namespace "prod" deleted
```

```
kunaltiwari@riot pr9 % █
```